

AfriSolve AI — Final Project Prototype

Summative Software Prototype: Complete Submission & Deployment Guide

Student Name: Chibuzor Moses Uzowuru
Course Deliverable: Final Project: Software Prototype [S:A] (Summative Assessment)
Target Total Score: 30 Points (System Requirements, Presentation, Code Availability, Deployment, Operation)

1. Assessment Rubric Breakdown (30 Points Total)

Source: Final Project_ Software Prototype [S_A].pdf. Full rubric marks require concrete proof of execution:

Criteria	Max Pts	Required Evidence for Top Marks (Band A)
Reflections of Requirements	10 pts	Captures all user functionalities highlighted in the SRS & system design (FR1–FR10 & BR1–BR8). Full problem !' vote !' verify !' AI !' project lifecycle.
Presentation	5 pts	Audible, clear 5–10 min video articulating problem, why it matters, proposed solution, and full interactive demo.
Code Availability	5 pts	Public GitHub repo; comprehensive step-by-step setup README allowing clean recreation.
Solution Deployment	5 pts	Live public URL accessible from anywhere on the internet (not localhost).
Operation	5 pts	Seamlessly functioning authentication, forms, buttons, redirections, and protected workflows.

Zero-Tolerance Warning on Submission Links

The course instructions warn: "Make sure all the links are working correctly and your google doc has access granted. Any incorrect links that end up not working or opening during grading will be graded Zero/0 points." Test all 4 deliverables in a signed-out browser window.

2. Step 1: Push Codebase to a Public GitHub Repository (5 Pts)

The local repository is initialized and committed on branch "main" (Commit e10d2a9). Follow these steps:

- **Create Repo:** Go to <https://github.com/new> and create a repository named "AfriSolve-AI".
- **Public Visibility:** Select PUBLIC. Leave "Add a README file", ".gitignore", and license UNCHECKED.
- **Link & Push:** Run the following commands in your terminal to link and push your code:

```
cd /home/wurucyber/AfriSolve-AI
git remote add origin https://github.com/<your-github-username>/AfriSolve-AI.git
git push -u origin main
```

- **Verify Signed-Out:** Open <https://github.com/<your-github-username>/AfriSolve-AI> in an Incognito/Private window. Confirm that the README renders cleanly and all source files are visible.

3. Step 2: Deploy to a Public URL (5 Pts)

The rubric requires a publicly accessible URL reachable over the internet. AfriSolve AI is packaged with a multi-stage Dockerfile and embedded PostgreSQL (PGLite).

Recommended Host: Render.com (Free & Zero-Config Docker)

Render provides free Docker web service hosting with persistent disk support:

- **Create Web Service:** Navigate to <https://dashboard.render.com> and click New + > Web Service.
- **Connect GitHub:** Connect your public GitHub repository "AfriSolve-AI".
- **Service Settings:** Configure Build & Runtime Settings:
 - Name: afrisolve-ai (or your preferred name)
 - Runtime: Docker (Render automatically detects the root Dockerfile)
 - Region: Choose closest region (e.g., Frankfurt or Oregon)
 - Instance Type: Free tier
- **Environment Variables:** Under Environment Variables, add the following production settings:

```
NODE_ENV=production
DEMO_MODE=false
SESSION_SECRET=c3f1b4a890e2d5c7f8a1b2c3d4e5f60718293a4b5c6d7e8f90a1b2c3d4e5f607
DATA_DIR=/app/data
```

- **Persistent Disk:** Add a Persistent Disk under the Disks tab. Name: "afrisolve_data", Mount Path: "/app/data", Size: 1 GB. This ensures that created user accounts, problem validations, and projects survive restarts.
- **Deploy:** Click Create Web Service. Render will build and deploy the container (approx. 2–3 minutes).
- **Initialize Grader Admin:** Bootstrap Grading Administrator Account. Once the service is live, open the Render "Shell" tab and run:

```
ADMIN_EMAIL=admin@afrisolve.org ADMIN_PASSWORD=SecureGrading2026! node server/bootstrap-admin.js
```

Alternative Host: Fly.io CLI Deployment

If you prefer Fly.io, run these commands from the project directory:

```
# Authenticate and launch container
fly launch --dockerfile Dockerfile --name afrisolve-ai
fly volumes create afrisolve_data --size 1
fly secrets set NODE_ENV=production DEMO_MODE=false
SESSION_SECRET=c3f1b4a890e2d5c7f8a1b2c3d4e5f60718293a4b5c6d7e8f90a1b2c3d4e5f607 DATA_DIR=/app/data
fly deploy
```

Testing Your Public Deployment

Open your deployed URL (e.g. <https://afrisolve-ai.onrender.com>) on your phone or in an incognito window. Verify that you can register a new account, sign in as the administrator, and browse problems.

4. Step 3: Record the 5–10 Minute Demo Video (5 Pts)

The assignment requires a self-recorded video clip between 5 and 10 minutes with your audible voice. Use OBS Studio, Open Screen, or SimpleScreenRecorder. Target 8 to 9 minutes following this exact cue sheet:

0:00–0:55 System & Problem: Introduce yourself (Chibuzor Moses Uzowuru). State problem: solution-first development in Africa where software fails due to lack of local evidence and community validation. Introduce AfriSolve AI.

0:55–1:40 Actors & Security: Highlight SRS/UML actors (Innovator, Researcher, Mentor, Admin). Show registration, demo mailbox preview (explaining real SMTP requirement), profile editor, and session security.

1:40–2:45 Problem Repository: Filter by 8 sectors and 14 countries. Submit a new evidence-bearing problem ("Unreliable cold storage for rural cooperative"). Show claim vs evidence separation and author edit ability.

2:45–3:40 Community Validation: Show two independent votes (BR3). Show that project creation is blocked before verification (BR5). Log in as Admin and approve/verify the problem (BR4).

3:40–4:35 AI Decision Support: Query the AI research module. Explain honest labelling of local heuristic fallback vs live Ollama. Highlight SDG mapping, literature links, and feasibility recommendations.

4:35–6:10 Project Collaboration: Create project from verified problem. Invite team member. Manage tasks on Kanban board (Todo/Doing/Done) and observe progress bar update. Upload file and view activity log.

6:10–7:05 Mentorship & Alerts: Search mentor directory. Submit mentorship request. Switch to mentor account to accept and add review feedback. Inspect in-app notification center.

7:05–8:00 Analytics & Admin: Show live database aggregate metrics. Download PDF and CSV reports. Switch to Admin: user activation toggles, audit logs, backup creation, and safe transactional restore.

8:00–8:45 Architecture & Close: Summarize React 19 + Express 5 + PGlite architecture. State academic boundaries. Conclude by referencing the 4 published links in your Google Doc.

Recording Recommendations

- Speak at a deliberate, confident pace.
- Rehearse once before recording.
- Upload the finished video to YouTube (Unlisted/Public) or Google Drive.
- Ensure Drive video permissions are set to "Anyone with the link can view".

5. Step 4: System Requirements & Verification (10 + 5 Pts)

AfriSolve AI fulfills all functional requirements (FR1–FR10) and business rules (BR1–BR8):

- **FR1: Discovery & Submission:** 8 sectors, 14 countries, SDG tags, evidence fields, author edit before verification.
- **FR2: Community Validation:** Multi-user independent voting, discussion comments, flag/report mechanism.
- **FR3: AI Decision Support:** Contextual research synthesis, SDG alignment, honest heuristic fallback labelling.
- **FR4: Project Workspace:** Verified-only creation, Kanban tasks, progress calculation, document attachments.
- **FR5: Mentorship Matching:** Expertise directory, structured requests, accept/decline, review feedback.
- **FR6: Notifications & Alerts:** In-app notification center, daily due-date reminders, demo mailbox preview.
- **FR7: Analytics & Reporting:** Real-time database aggregates, sector/country distributions, PDF & CSV export.
- **FR8: Administration & Audit:** Role governance, account activation, content review queue, immutable audit logs.
- **FR9: Automated Backups:** Daily automated schedule, manual snapshots, safe transactional restore.
- **FR10: User Profile & Security:** Email verification, script password hashing, HTTP-only cookies, base64 photo validation.

Automated Test Suite Evidence (100% Passed)

- Backend API Tests: 13/13 passed (node --test tests/api.test.js tests/backend-*.test.js)
- UI Vitest Tests: 12/12 passed (vitest run)
- Playwright E2E Tests: 3/3 passed (full browser navigation, lifecycle, and 390px mobile viewport)
- Vite Client Build: Clean production build in 1.04s

6. Step 5: Prepare Google Doc Deliverable & 4 Mandatory Links

Create a Google Document titled with the required name structure:

```
Chibuzor_Moses_Uzowuru_[Summative]_[09302026]
```

Include the four verified links in the document:

- **1. Demonstration Video URL:** <https://youtu.be/your-video-id> (or Google Drive share link)
- **2. Public GitHub Repository URL:** <https://github.com/<your-username>/AfriSolve-AI>
- **3. SRS Document Link:** <https://drive.google.com/file/d/your-srs-doc-id/view?usp=sharing>
- **4. Public Application URL:** <https://afrisolve-ai.onrender.com> (or your hosting URL)

7. Final Pre-Submission Inspection Checklist

- ☐ Google Doc sharing set to "Anyone with the link can view".
- ☐ GitHub repository visibility is PUBLIC (tested in an incognito window).
- ☐ GitHub README contains complete step-by-step setup instructions.
- ☐ Deployed URL is live, reachable over HTTPS, and accessible from an outside device.
- ☐ Grading administrator account initialized on the live deployment.
- ☐ Demonstration video is 5–10 minutes, audible, and plays without requiring login.
- ☐ SRS document link opens to Assignment 1 / 2 without permission errors.
- ☐ All 4 links in the Google Doc have been tested and verified in a private window.